Physical Chemistry of Materials

Lecture 04

Foundations of Statistical Mechanics

Last Class

Two Systems in Equilibrium

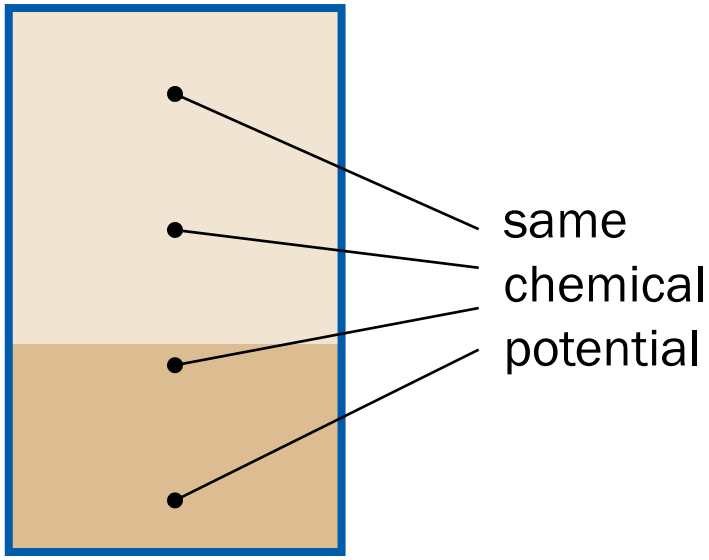
$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V + \mu\mathrm{d}n$$

- Mechanistic equilibrium: $p_1 = p_2$
  - Conjugate extensive variable: $\mathrm{d}V$
- Thermal equilibrium: $T_1 = T_2$
  - Conjugate extensive variable: $\mathrm{d}S = \delta q/T$
- Diffusive equilibrium: $\mu_1 = \mu_2$
  - Conjugate extensive variable: $\mathrm{d}n$

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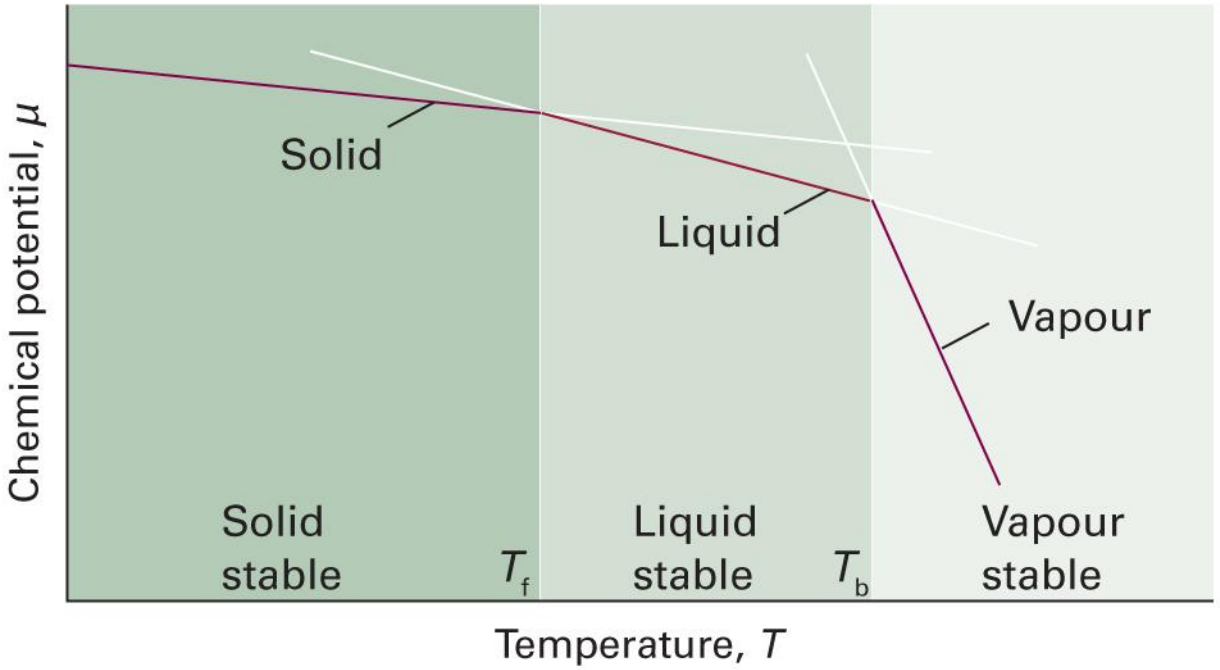
Thermodynamics of Phases

At equilibrium, the chemical potential of a substance is the same in and throughout every phase in the system



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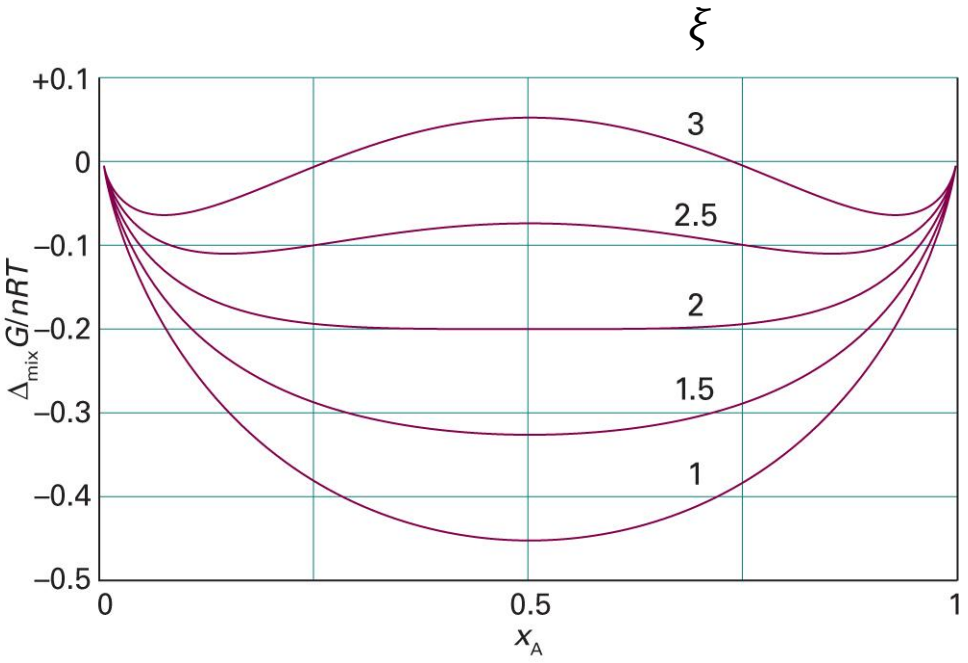
Chemical Potential



(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 12th ed., Figures 4B.1)

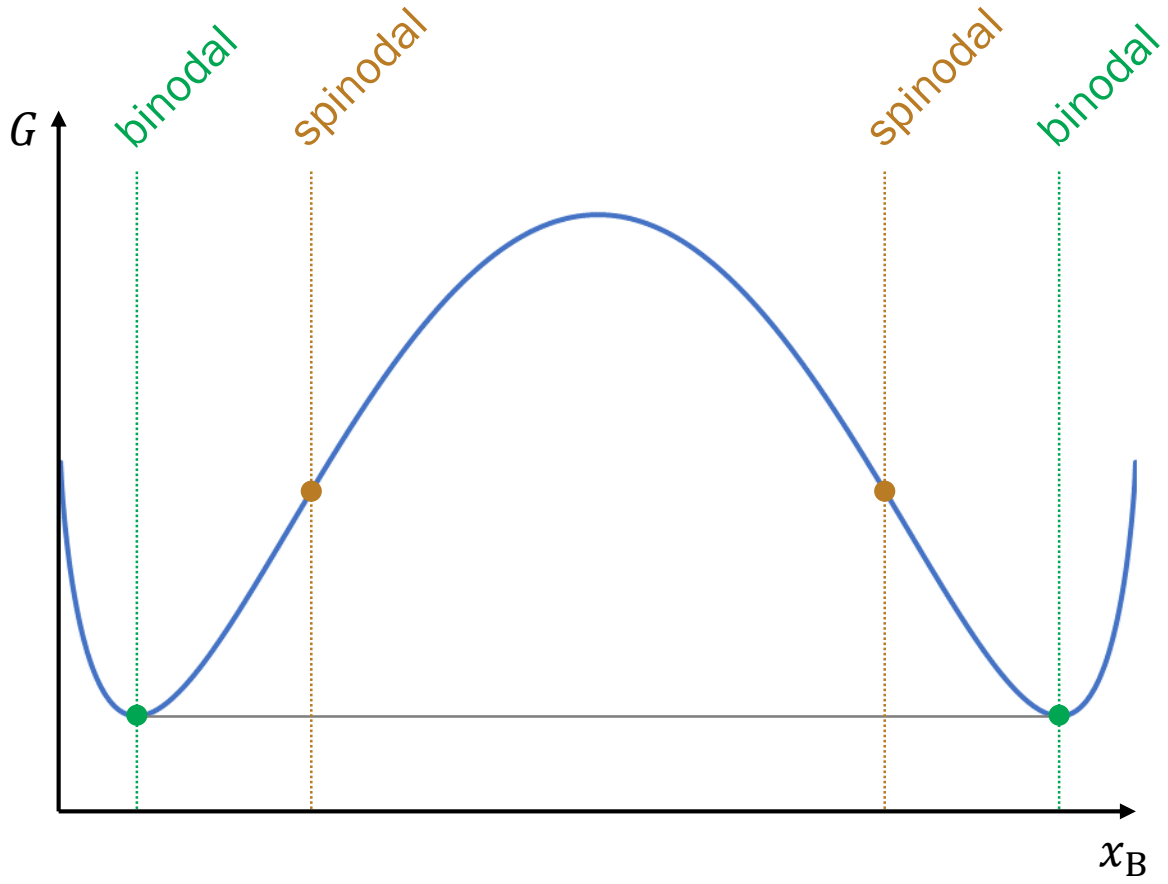
Gibbs Energy of Mixing

$$\Delta_{\text{mix}}G = nRT(x_{\text{A}} \ln x_{\text{A}} + x_{\text{B}} \ln x_{\text{B}} + \xi x_{\text{A}} x_{\text{B}})$$



(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 12th ed., Figure 5B.5)

Binodal and Spinodal



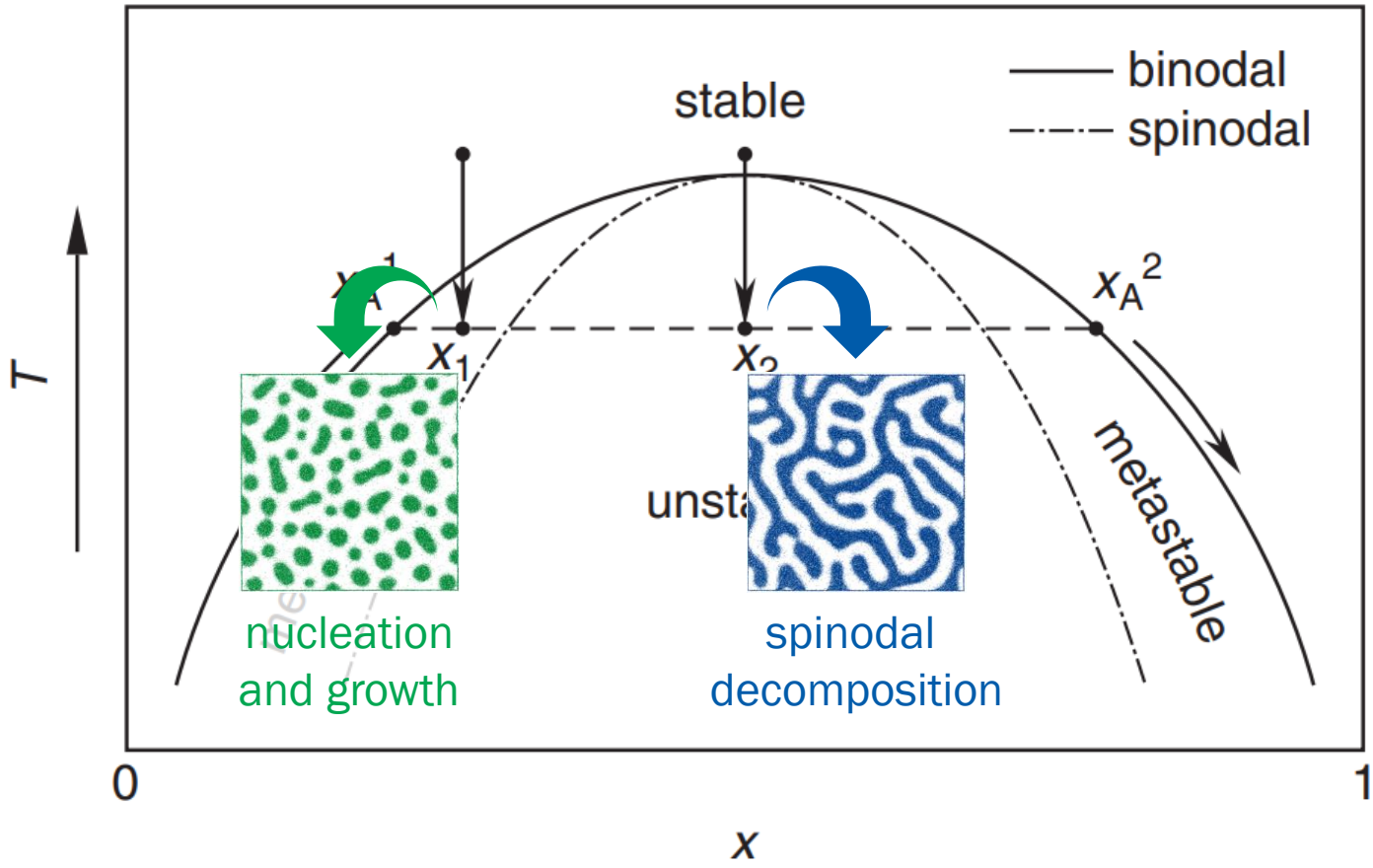
Binodal:

$$\frac{\partial G}{\partial x_B} = 0$$

Spinodal:

$$\frac{\partial^2 G}{\partial x_B^2} = 0$$

Kinetics of Phase Separation

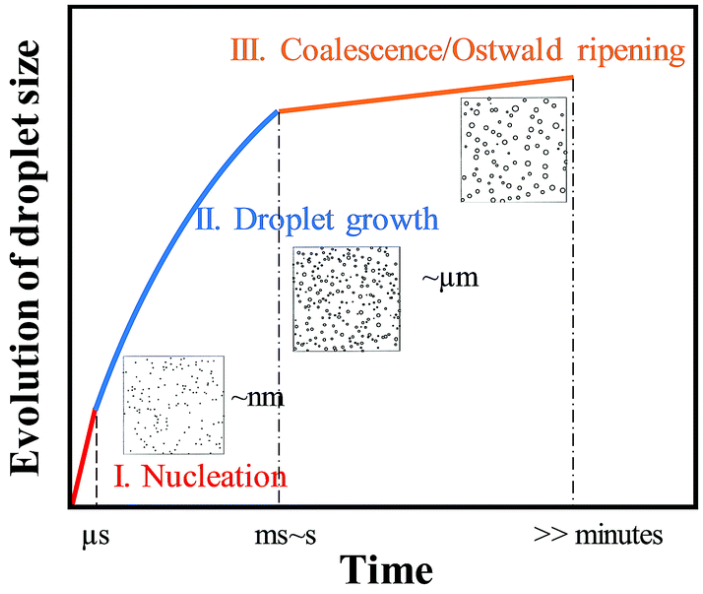


(Images: Cheng, ***Phase Transitions in Polymers*** (2008), Fig. 2.14 / Gebauer *et al.*, ***Chem Soc Rev*** (2014), Fig. 3)

Last Class

Nucleation and Growth

metastable phase ⟶ stable phase



Universal in chemistry and physics!

(Image: Luo *et al.*, ***Phys Chem Chem Phys*** (2017), Fig. 6)

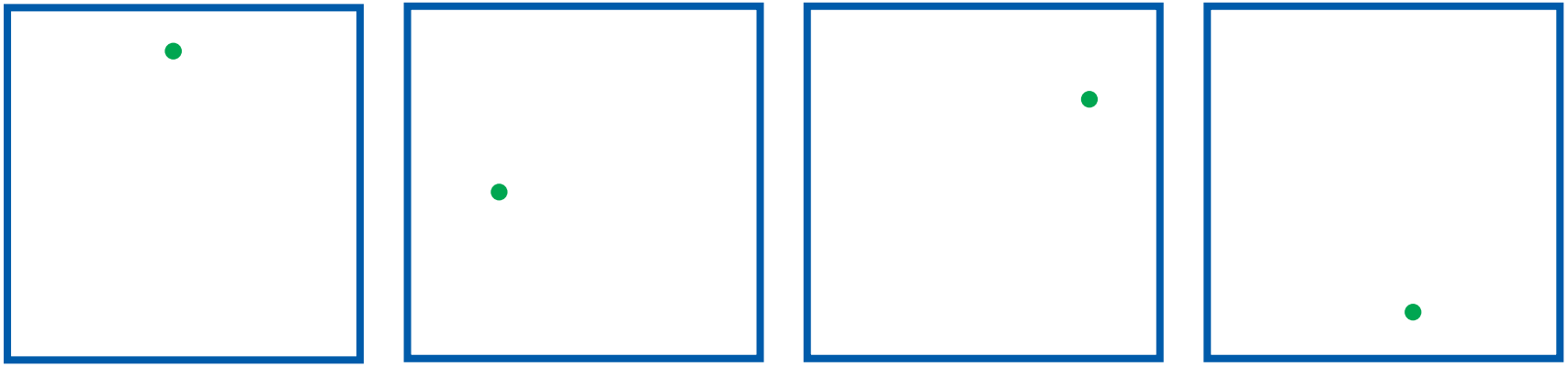
“States”

- (Macro)state: defined by values of macroscopic variables
- Microstate: specific microscopic arrangement
- Different microstates can be mapped into the same macrostate

Example: Diffusion

When a gas molecule moves around the system,

- Macrostate: defined by the volume V
- Microstate: the position of the molecule



(Thermodynamic) Ensemble

Idealized group of a large number of virtual copies

(sometimes infinitely many) of a system

Microcanonical ensemble

Canonical ensemble

Grand canonical ensemble

⋮

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Microcanonical Ensemble

Thermodynamic ensemble of an isolated system

- Conserved quantities: N, V, U (or traditionally N, V, E)

Fundamental Assumption of SM

"In an isolated system in equilibrium,

all accessible microstates are equally probable."

Microcanonical Ensemble

Thermodynamic ensemble of an isolated system

- Conserved quantities: N, V, U (or traditionally N, V, E)
- All microstates are equally probable
- Probability to find the system in microstate j

$$p_j = \frac{1}{W}$$

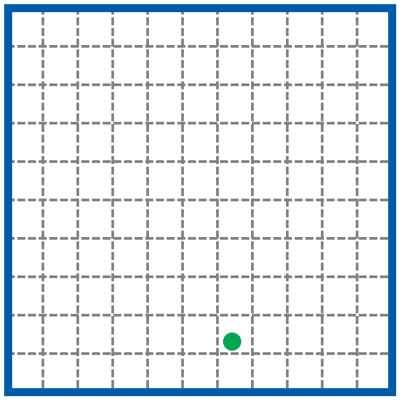
Statistical Definition of Entropy

- W = number of microstates belonging to a specific macrostate (multiplicity, degeneracy)

- Statistical definition of entropy (for an isolated system in equilibrium)

$$S = k_{\mathrm{B}} \ln W$$

Example: Diffusion



Introduce the “unit volume” v_0.

For the system volume V, $W(V) = V/v_0$

Example: Diffusion

Suppose that the system volume changes from V_i to V_f:

$$W_\mathrm{i} = V_\mathrm{i}/v_0$$

$$W_\mathrm{f} = V_\mathrm{f}/v_0$$

The change in entropy is given by

$$\Delta S_1 = k_\mathrm{B} \ln(W_\mathrm{f}/W_\mathrm{i}) = k_\mathrm{B} \ln(V_\mathrm{f}/V_\mathrm{i})$$

For an ideal gas consisting of N molecules,

$$\Delta S_N = N k_\mathrm{B} \ln(V_\mathrm{f}/V_\mathrm{i}) = nR \ln(V_\mathrm{f}/V_\mathrm{i})$$

(Thermodynamic) Ensemble

Idealized group of a large number of virtual copies

(sometimes infinitely many) of a system

Microcanonical ensemble

Canonical ensemble

Grand canonical ensemble

⋮

Canonical Ensemble

Thermodynamic ensemble of a closed system

- Conserved quantities: N, V, T
- Microstates may have different energies
- All microstates are NOT equally probable
- Probability to find the system in some microstate?

Shannon Entropy

Consider a system with the following properties:

- t microstates
- each microstate i with probability p_i
- Ensemble: $\mathcal{N}$ independent replicates ($\mathcal{N} \to \infty$)

Among $\mathcal{N}$, the number of finding microstate i is $n_i = \mathcal{N} p_i$

Then, the number of combinations is

$$W = \frac{\mathcal{N}!}{n_1! \, n_2! \cdots n_t!}$$

Stirling's Approximation

$$\ln N! \approx N \ln N - N \text{ (for large } N\text{)}$$

$$\begin{aligned}
\ln N! &= \ln\{N \times (N-1) \times (N-2) \times \cdots\} \\
&= \ln N + \ln(N-1) + \ln(N-2) + \cdots \\
&= \sum_{k=1}^{N} \ln k \\
&\approx \int_1^N \ln x \, dx \\
&= [x \ln x - x]_1^N \\
&= N \ln N - N + 1 \\
&\approx N \ln N - N \text{ (for large } N \gg 1\text{)}
\end{aligned}$$

Shannon Entropy

$$W = \frac{\mathcal{N}!}{n_1!\, n_2! \cdots n_t!}$$

$$S = k_\mathrm{B} \ln W = k_\mathrm{B} \ln \frac{\mathcal{N}!}{n_1!\, n_2! \cdots n_t!}$$

By Stirling's approximation,

$$S \approx k_\mathrm{B}(\mathcal{N} \ln \mathcal{N} - \textstyle\sum_i n_i \ln n_i) = -k_\mathrm{B} \textstyle\sum_i n_i \ln(n_i/\mathcal{N})$$

$$S = -\mathcal{N} k_\mathrm{B} \textstyle\sum_i p_i \ln p_i$$

Hence, "average" entropy per system is $S = -k_\mathrm{B} \sum_i p_i \ln p_i$

Entropy and Information



$p_1 = 1 \quad p_2 = 0 \quad p_3 = 0 \quad p_4 = 0 \quad p_5 = 0 \quad p_6 = 0 \quad \ldots \quad p_t = 0$

$S = 0$ (minimum)



$p_1 = \frac{1}{t} \quad p_2 = \frac{1}{t} \quad p_3 = \frac{1}{t} \quad p_4 = \frac{1}{t} \quad p_5 = \frac{1}{t} \quad p_6 = \frac{1}{t} \quad \ldots \quad p_t = \frac{1}{t}$

$S =$ maximum

Principle of Maximum Entropy

The probability distribution which best represents the current state of knowledge about a system is the one with largest entropy

Maximize Shannon Entropy

$$S/k_{\mathrm{B}} = -\sum_i p_i \ln p_i$$

We always have the normalization constraint, $\sum_i p_i - 1 = 0$.

Apply Lagrange's multiplier method:

$$\frac{\partial[S/k_{\mathrm{B}} - \lambda(\sum_i p_i - 1)]}{\partial p_i} = -(\ln p_i + 1) - \lambda = 0$$

$$p_i = e^{-1-\lambda}$$

Using the normalization constraint, $p_i = 1/W$

(microcanonical ensemble)

Ensemble Energy Conservation

$$S/k_{\mathrm{B}} = -\sum_i p_i \ln p_i$$

$$\sum_i p_i - 1 = 0$$

We can introduce one more constraint: $\sum_i n_i E_i = \mathcal{N} U$

As the constraint implies $\sum_i p_i E_i - U = 0$,

$$\frac{\partial[S/k_{\mathrm{B}} - \lambda(\sum_i p_i - 1) - \beta(\sum_i p_i E_i - U)]}{\partial p_i} = 0$$

$$p_i = e^{-1-\lambda-\beta E_i}$$

Partition Function

$$p_i = e^{-1-\lambda-\beta E_i}$$

From the normalization constraint,

$$\sum_i p_i = e^{-1-\lambda} \sum_i e^{-\beta E_i} = 1$$

Define the partition function Z:

$$Z = e^{1+\lambda} = \sum_i \underbrace{e^{-\beta E_i}}_{\text{Boltzmann factor}}$$

Back to Entropy

$$p_i = e^{-\beta E_i}/Z$$

$$S/k_\mathrm{B} = -\sum_i p_i \ln p_i = -\sum_i p_i \ln\left(\frac{e^{-\beta E_i}}{Z}\right)$$

$$S/k_\mathrm{B} = -\sum_i p_i(-\beta E_i - \ln Z) = \beta \sum_i p_i E_i + \ln Z$$

$$\therefore \; S/k_\mathrm{B} = \beta U + \ln Z$$

Bridge to Thermodynamics

$$S = k_{\mathrm{B}}\beta U + k_{\mathrm{B}} \ln Z$$

From thermodynamics, we know

$$\left(\frac{\partial S}{\partial U}\right)_{V,n} = \frac{1}{T}$$

Hence, $k_{\mathrm{B}}\beta = 1/T$ or $\beta = 1/k_{\mathrm{B}}T$ (inverse temperature)

Also,

$$-k_{\mathrm{B}}T \ln Z = U - TS = A \text{ (Helmholtz energy)}$$

Summary

For a canonical ensemble, the probability to find the system in a microstate i (with energy E_i) is

$$p_i = e^{-1-\lambda-\beta E_i} = \frac{e^{-\beta E_i}}{Z} = \frac{e^{-E_i/k_\mathrm{B}T}}{Z}$$

where the partition function $Z = \sum_i e^{-\beta E_i} = \sum_i e^{-E_i/k_\mathrm{B}T}$

Also, the Helmholtz energy is given by

$$A = -k_\mathrm{B}T \ln Z$$

Derived Functions

$$A = -k_{\mathrm{B}} T \ln Z$$

Using $\mathrm{d}A = -S\mathrm{d}T - p\mathrm{d}V + \mu\mathrm{d}n$,

$$S = -\left(\frac{\partial A}{\partial T}\right)_{V,n}, \qquad p = -\left(\frac{\partial A}{\partial V}\right)_{T,n}, \qquad \mu = \left(\frac{\partial A}{\partial n}\right)_{T,V}$$

One More Useful Relation

$$U = \sum_i p_i E_i = \sum_i \frac{E_i e^{-\beta E_i}}{Z}$$

However,

$$\frac{\partial \ln Z}{\partial \beta} = \frac{1}{Z}\frac{\partial Z}{\partial \beta} = -\frac{1}{Z}\sum_i E_i e^{-\beta E_i}$$

Hence,

$$U = -\frac{\partial \ln Z}{\partial \beta}$$

Partition Function and …

- Helmholtz energy: $A = -k_{\mathrm{B}} T \ln Z$
- Internal energy: $U = -\partial(\ln Z)/\partial\beta$
- Entropy: $S = (U - A)/T$

"A partition function contains

all the thermodynamic information about a system."

Z and Thermodynamics

